CH11. Metal-Oxide-Semiconductor Field-Effect Transistor: Additional Concepts

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Glossary of Important Terms

Term	Definition
channel length modulation	The change in effective channel length with drain-to-source voltage when the MOSFET is biased in saturation.
drain-induced barrier lowering	The near punch-through condition in which the potential barrier between the source and channel region in an off transistor is lowered due to a large applied drain voltage.
hot electrons	Electrons with energies far greater than the thermal-equilibrium value caused by acceleration in high electric fields.
lightly doped drain (LDD)	A MOSFET with a lightly doped drain region adjacent to the channel to reduce voltage breakdown effects.
narrow-channel effects	The shift in threshold voltage as the channel width narrows.
near punch-through	The reduction in the potential barrier between source and substrate by the drain-to-substrate voltage, resulting in a rapid increase in drain current.
short-channel effects	The shift in threshold voltage as the channel length becomes smaller.
snapback	The negative resistance effect during breakdown in a MOSFET caused by the variable current gain in a parasitic bipolar transistor.
subthreshold conduction	The process of current conduction in a MOSFET when the transistor is biased below the threshold inversion point.
surface scattering	The process of electric field attraction and coulomb repulsion of carriers at the oxide–semiconductor interface as the carriers drift between source and drain.
threshold adjustment	The process of altering the threshold voltage by changing the semiconductor doping concentration through ion implantation.

주요 용어 정리

용어	정의
채널 길이 변조 channel length modulation	MOSFET이 포화 영역에 바이어스되었을 때, 드레인-소스 전압에 따라 유효 채널 길이가 변하는 현상
드레인 유도 장벽 저하 drain-induced barrier lowering	큰 드레인 전압이 인가되어, off 상태 트랜지스터에서 소스와 채널 영역 사이의 전위 장벽이 낮아지는 준펀치스루 상태
핫 전자 hot electrons	높은 전계에 의해 가속되어 열평형 값보다 훨씬 큰 에너지를 갖게 된 전자
저농도 도핑 드레인 lightly doped drain, LDD	전압 항복 효과를 줄이기 위해 채널에 인접한 드레인 영역을 저농도로 도핑한 MOSFET
좁은 채널 효과 narrow-channel effects	채널 폭이 좁아짐에 따라 문턱 전압이 변하는 현상
준펀치스루 near punch-through	드레인-기판 전압에 의해 소스와 기판 사이의 전위 장벽이 낮아져 드레인 전류가 급격히 증가하는 현상
짧은 채널 효과 short-channel effects	채널 길이가 짧아짐에 따라 문턱 전압이 변하는 현상
스냅백 snapback	기생 바이폴라 트랜지스터의 가변 전류 이득으로 인해 MOSFET 항복 시 나타나는 음성 저항 효과
문턱전압 이하 전도 subthreshold conduction	트랜지스터가 문턱 반전점 아래로 바이어스되었을 때 MOSFET에 전류가 흐르는 과정
표면 산란 surface scattering	캐리어가 소스와 드레인 사이를 드리프트할 때 산화막-반도체 계면에서 전계 인력과 쿨롱 반발을 받는 과정
문턱 전압 조절 threshold adjustment	이온 주입으로 반도체 도핑 농도를 바꾸어 문턱 전압을 조절하는 과정

11.0 Preview

1. Analyze sub threshold conduction
2. Channel length modulation.
   - Short channel lengths and leads to a finite output resistance.
3. Carrier mobility degradation due to high gate field.
4. The effects of carrier saturation velocity.
   - Carriers can easily reach their saturation velocity in short-channel devices.
5. Discuss MOSFET scaling
   - How various parameters must be changed.
   - Device size is decreased.
6. Deviation in V_{th} due to small geometry devices.
   - Short channel length and width.
7. Voltage breakdown mechanisms in MOSFETs
8. Vth adjustment by ion implantation.
9. The introduction of trapped oxide charges by ionizing radiation and hot electron effects.

11.1.1 Subthreshold Conduction

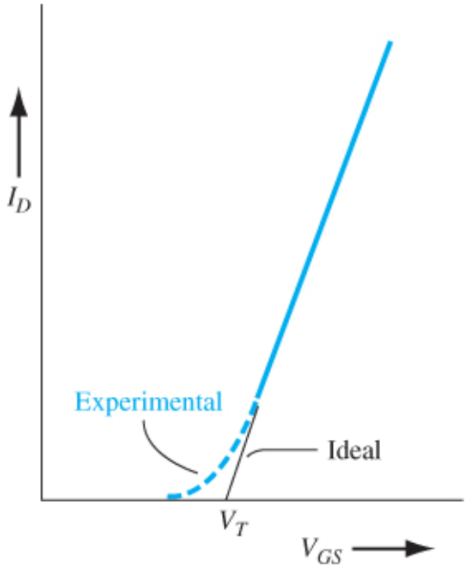


Figure 11.1 | Comparison of ideal and experimental plots of $\sqrt{I_D}$ versus V_{GS}.

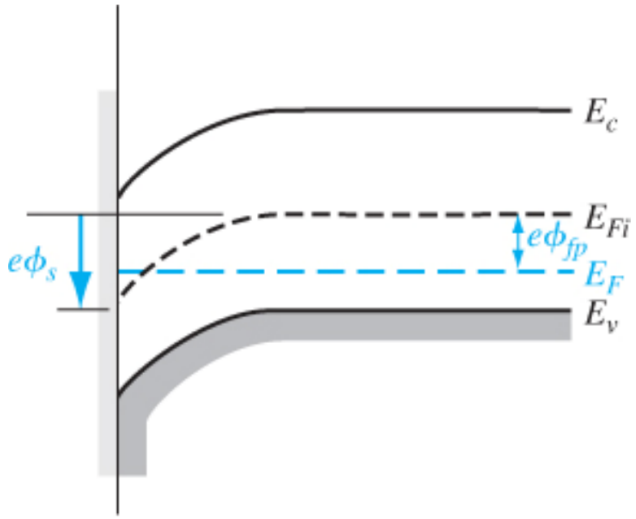


Figure 11.2 | Energy-band diagram when $\phi_{fp} < \phi_s < 2\phi_{fp}$.

1. Ideal drain current should be zero when V_{GS} is less than V_{th}.
2. Experimentally, I_{DS} is not zero when $V_{GS} < V_{th}$.
3. The drain current, which exists for $V_{GS} < V_{th}$, is known as the subthreshold current.

11.1.1 Subthreshold Conduction

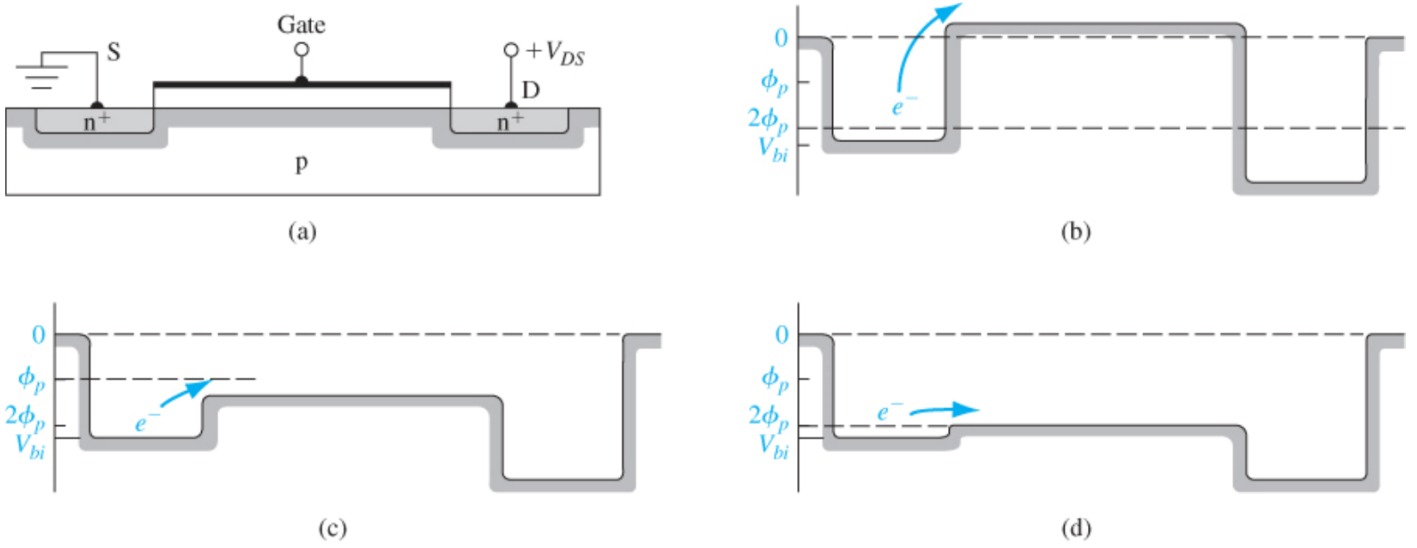


Figure 11.3 | (a) Cross section along channel length of n-channel MOSFET. Energy-band diagrams along channel length at (b) accumulation, (c) weak inversion, and (d) inversion.

1. Some conduction between the n^+ source and drain contacts through this weakly inverted channel ($\phi_{fp} < \phi_s < 2\phi_{fp}$).
2. Fig.11.3➔ the surface potential along the length of the channel at
   (1) Accumulation, (2) Weak inversion, (3) Inversion.
   (2) V_{DS} is very small (~0V)

11.1.1 Subthreshold Conduction

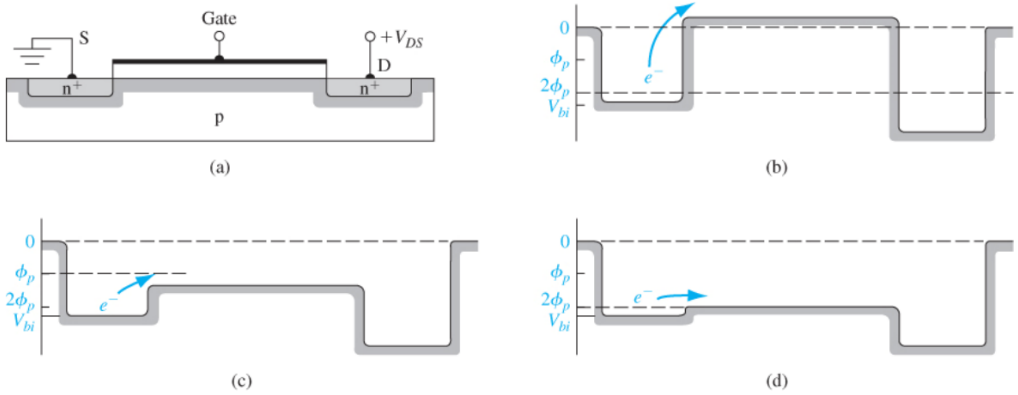


Figure 11.3 | (a) Cross section along channel length of n-channel MOSFET. Energy-band diagrams along channel length at (b) accumulation, (c) weak inversion, and (d) inversion.

1. Potential barrier between the n^+ source and channel region which the electrons must overcome in order to generate a channel current.
2. (d) In the inversion mode, the barrier is so small.➔ "loss of exponential dependence." since the junction is more like an ohmic contact.
3. If V_{DS} is larger than a few (kT/e) volts, the subthreshold current is independent of V_{DS}.

$$I_D(sub) \propto \left[\exp\left(\frac{eV_{GS}}{kT}\right)\right] \cdot \left[1 - \exp\left(\frac{-eV_{DS}}{kT}\right)\right] \quad (11.1)$$

11.1.1 Subthreshold Conduction

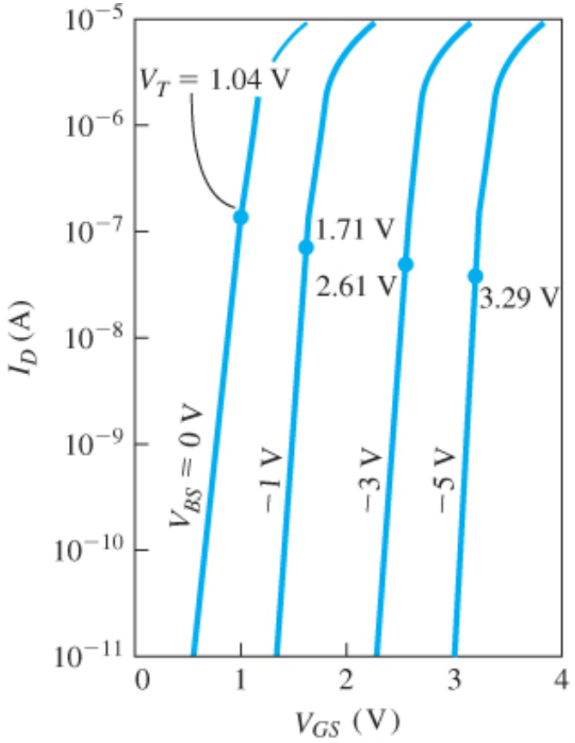


Figure 11.4 | Subthreshold current–voltage characteristics for several values of substrate voltage (the threshold voltage is indicated on each curve). *(From Schroder [17].)*

$$I_D(sub) \propto \left[\exp\left(\frac{eV_{GS}}{kT}\right)\right] \cdot \left[1-\exp\left(\frac{-eV_{DS}}{kT}\right)\right] \quad (11.1)$$

1. Exponential behavior of the subthreshold current for several body-to-source votages.
2. V_{GS} change on approximately 60 mV➔ an order of magnitude change in the subthreshold current.
3. Detailed analysis:
   - The slope of the I_{DS} versus V_{DS} curve
   - A function of the semiconductor doping.
   - A function of the interface state density.
   - The measurement of the slope of these curves ➔ the oxide-semiconductor interface state density.
   - The subthreshold current may add significantly to power dissipation in a large-scale integrated circuits which hundreds of thousands of MOSFETs.
   - The circuit design must include the I_{sub}.

11.1.2 Channel Length Modulation

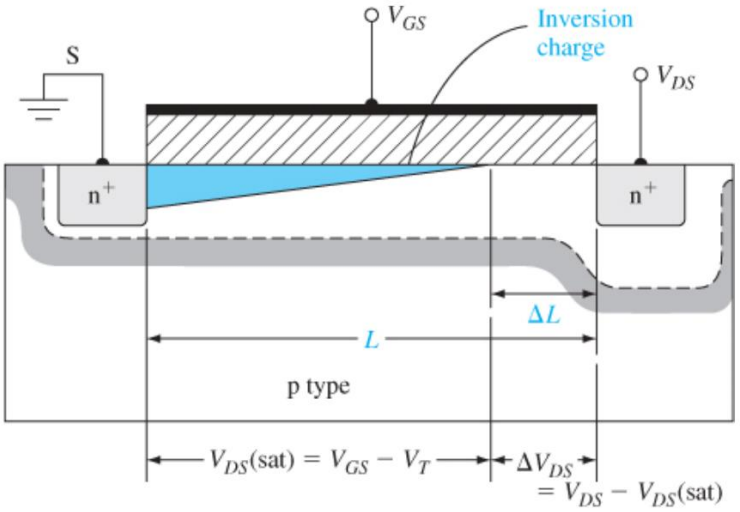


Figure 11.5 | Cross-section of an n-channel MOSFET showing the channel length modulation effect.

$$x_p = \sqrt{\frac{2\varepsilon_s \phi_{fp}}{eN_a}} \quad (11.2)$$

$$x_p = \sqrt{\frac{2\varepsilon_s}{eN_a}\left(\phi_{fp} + V_{DS}\right)} \quad (11.3)$$

1. Assumption: Channel length L was a constant.
2. MOSFETs in the saturation region, the depletion region at the drain terminal extends laterally into the channel, reducing the effective channel length.
3. Since the depletion region width is bias dependent, the effective channel length is also bias dependent and is modulated by the drain-to-source voltage.

11.1.2 Channel Length Modulation

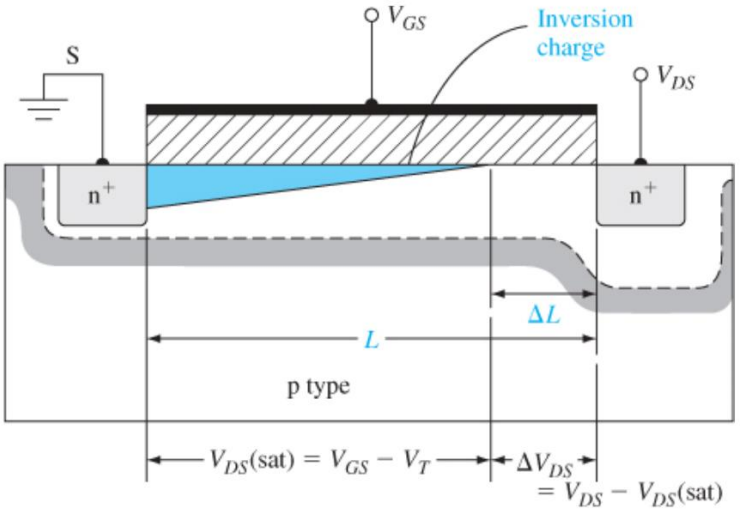


Figure 11.5 | Cross-section of an n-channel MOSFET showing the channel length modulation effect.

1. The applied drain-to-source voltage is V_{DS} and we are assuming that $V_{DS} > V_{DS(sat)}$.
2. Second approximation at determining ΔL, one-dimensional Poisson's equation.
3. Electric field E_{sat} is the lateral electric field at the point where the inversion layer charge is pinched off.

$$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\phi_{fp} + V_{DS}(sat) + \Delta V_{DS}} - \sqrt{\phi_{fp} + V_{DS}(sat)}\right] \quad (11.4)$$

$$\Delta V_{DS} = V_{DS} - V_{DS}(sat) \quad (11.5)$$

11.1.2 Channel Length Modulation

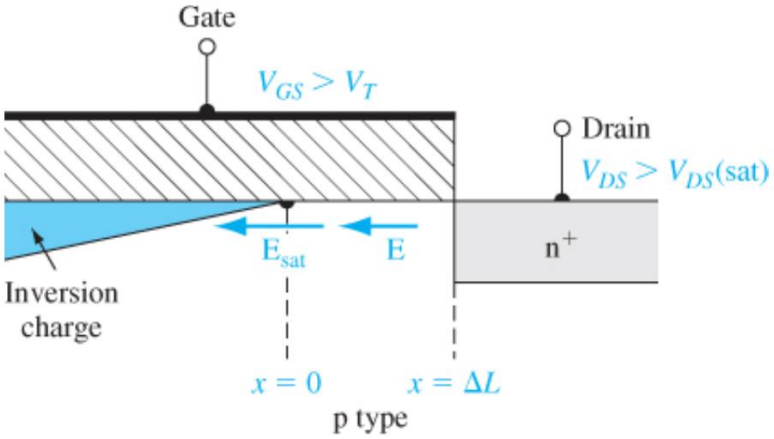


Figure 11.6 | Expanded view of cross section near the drain terminal of an n-channel MOSFET showing the channel length modulation effect.

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon_s} \quad (11.6)$$

$$E = -\frac{eN_a x}{\varepsilon_s} - E_{sat} \quad (11.7)$$

$$\phi(x) = -\int E dx = \frac{eN_a x^2}{2\varepsilon_s} + E_{sat} x + C_1 \quad (11.8)$$

$$V_{DS} = \frac{eN_a(\Delta L)^2}{2\varepsilon_s} + E_{sat}(\Delta L) + V_{DS}(sat) \quad (11.9)$$

(boundary conditions $\phi(x=0)=V_{DS}(sat)$ and $\phi(x=\Delta L)=V_{DS}$)

Solving for ΔL, we can write; $$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\phi_{sat} + [V_{DS} - V_{DS}(sat)]} - \sqrt{\phi_{sat}}\right] \quad (11.10)$$

11.1.2 Channel Length Modulation

$$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\phi_{sat} + \left[V_{DS} - V_{DS}(sat)\right]} - \sqrt{\phi_{sat}}\right] \quad (11.10)$$

Where

$$\phi_{sat} = \frac{2\varepsilon_s}{eN_a} \cdot \left(\frac{E_{sat}}{2}\right)$$

In general, the value of Esat is in the range $10^4 < E_{sat} < 2\times10^5$ V/cm

Other models used to determine ΔL include the negative charges due to the drain current. Also, include two-dimensional effects. These models are not considered here.

$$I_D' = \left(\frac{L}{L - \Delta L}\right) I_D \quad (11.11)$$

Where I_D' is the actual current and I_D is the ideal drain current.

ΔL is a function of V_{DS}, I_D is now also function of VDS even though the transistor is biased in the saturation region.

11.1.2 Channel Length Modulation

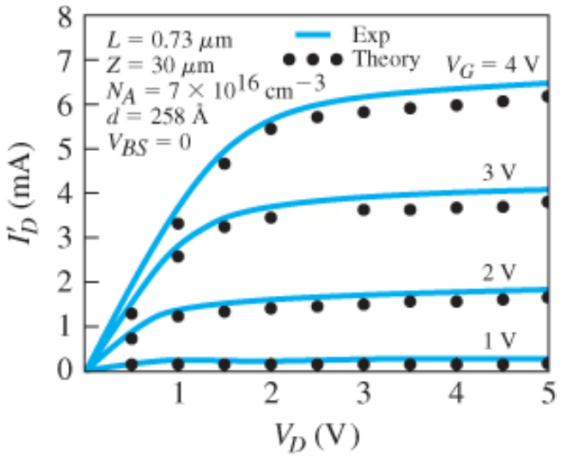


Figure 11.7 | Current–voltage characteristics of a MOSFET showing short-channel effects. *(From Sze [22].)*

$$I_D' = \frac{k_n'}{2} \cdot \frac{W}{L} \cdot \left[(V_{GS} - V_T)^2 (1 + \lambda V_{DS}) \right] \quad (11.12)$$

λ: the channel length modulation parameter.

$$r_o = \left(\frac{\partial I_D'}{\partial V_{DS}} \right)^{-1} = \left\{ \frac{k_n'}{2} \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \cdot \lambda \right\}^{-1} \quad (11.13a)$$

Since λ is normally small, Equation (11.13a) can be written as .

$$r_o \cong \frac{1}{\lambda I_D} \quad (11.13b)$$

Example 11.1(1)

① **Determine the increase in drain current due to short channel modulation.**
- **N_a=2×10^{16} cm^{-3}, V_T=0.4 V, L=1 μm, V_{GS}=1 V, V_{DS}=2.5 V**

We find

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{ni}\right) = (0.0259)\ln\left(\frac{2\times10^{16}}{1.5\times10^{10}}\right) = 0.3653V$$

$$V_{DS}(sat) = V_{GS} - V_T = 1.0 - 0.4 = 0.6V$$

and

$$\Delta V_{DS} = V_{DS} - V_{DS}(sat) = 2.5 - 0.6 = 1.9V$$

Using Equation (11.4), we determine

$$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\phi_{fp} + V_{DS}(sat) + \Delta V_{DS}} - \sqrt{\phi_{fp} + V_{DS}(sat)}\right]$$

$$= \sqrt{\frac{2(11.7)(8.85\times10^{-14})}{(1.6\times10^{-19})(2\times10^{16})}}\left[\sqrt{0.3653+0.6+1.9} - \sqrt{0.3653+0.6}\right]$$

$$= 1.807\times10^{-5}\,cm$$

or

$$\Delta L = 0.1807\,\mu m$$

Example 11.1(2)

① Determine the increase in drain current due to short channel modulation.
- $N_a=2\times10^{16}$ cm^{-3}, $V_T=0.4$ V, $L=1$ μm, $V_{GS}=1$ V, $V_{DS}=2.5$ V

Then

$$\frac{I'_D}{I_D}=\frac{L}{L-\Delta L}=\frac{1}{1-0.1807}=1.22$$

- The actual drain current increases as the effective channel length decreases when the transistor is biased in the saturation region.

11.1.3 Mobility Modulation

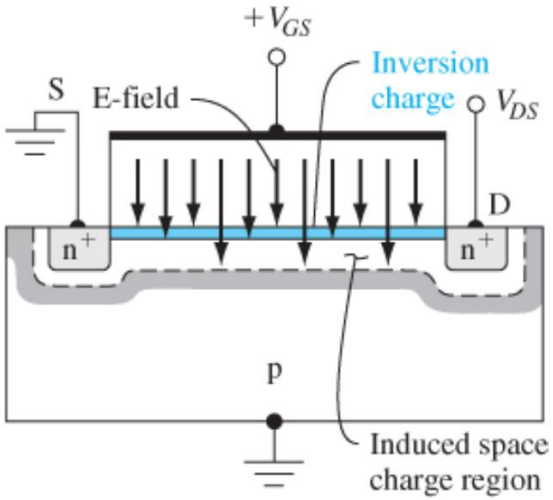


Figure 11.8 | Vertical electric field in an n-channel MOSFET.

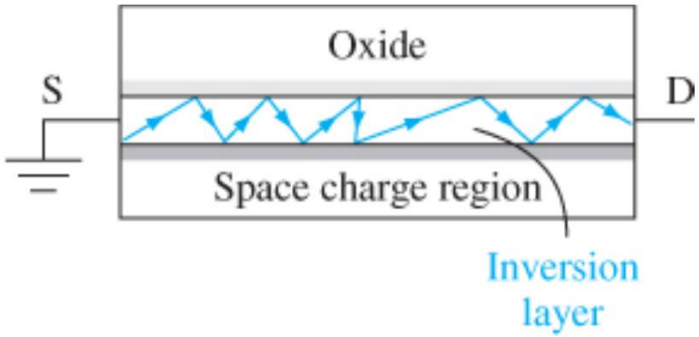


Figure 11.9 | Schematic of carrier surface scattering effects.

$$E_{eff} = \frac{1}{\varepsilon_s}\left(\left|Q'_{SD}(\max)\right| + \frac{1}{2}Q'_n\right) \quad (11.14)$$

The effective inversion charge mobility can be determined from the channel conductance at a function of gate voltage.

11.1.3 Mobility Modulation

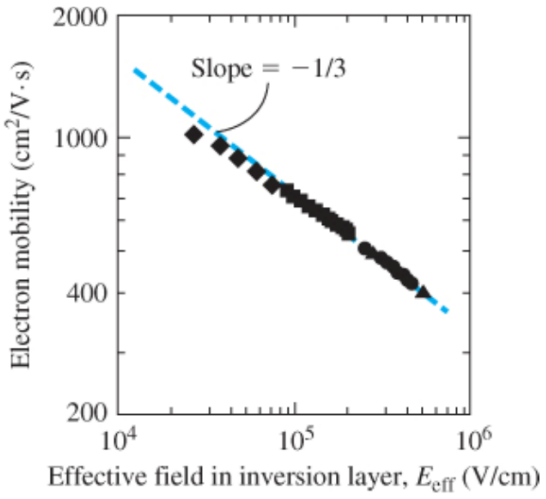


Figure 11.10 | Measured inversion layer electron mobility versus electric field at the inversion layer.
(From Yang [25].)

The effective inversion charge mobility Function of temperature because of lattice scattering. As the temperature is reduced, the mobility increases.

The effective electron mobility at T=300K for different doping levels and different oxide thicknesses.

The effective mobility is only a function of the electric field a the inversion layer and is independent of oxide thickness.

The effective mobility may be represented by

$$\mu_{eff} = \mu_0 \left(\frac{E_{eff}}{E_0} \right)^{-\frac{1}{3}} \qquad (11.15)$$

Where μ_o and E_o are constants determined from experimental results.

Example 11.2

① **Calculate the effective electric field at threshold for a given semiconductor doping concentration.**
- **T=300 K, N_a=3×10^{16} cm^{-3}**

From the results of Capter 10, we can calculate

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{ni}\right) = (0.0259)\ln\left(\frac{3\times10^{16}}{1.5\times10^{10}}\right) = 0.376V$$

and

$$\chi_{dT} = \left\{\frac{4\varepsilon_s\phi_{fp}}{eN_a}\right\}^{1/2} = \left\{\frac{4(11.7)(8.85\times10^{-14})(0.376)}{(1.6\times10^{-19})(3\times10^{16})}\right\}^{1/2}$$

At the threshold inversion point, we may assume that $Q'_n = 0$, so the erffective electirc field from Equation (11.14) is found as

$$E_{eff} = \frac{1}{\varepsilon s}\left|Q'_{SD}(\max)\right| = \frac{8.64\times10^{-8}}{(11.7)(8.85\times10^{-14})} = 8.34\times10^4 V/cm$$

- **We can see, from Figure 11.10, that this value of effective transverse electric field at the surface is sufficient for the effective inversion charge mobility to be significantly less than the bulk semiconductor value.**

11.1.4 Velocity Saturation

Assumption:

(1) For long channel device, the mobility to be constant, which means that the drift velocity increases without limit as the electric field increases.

(2) In the ideal I-V relationship, current saturation occurs when the inversion charge density becomes zero at the drain terminal;

(3) $V_{DS}=V_D(sat)=V_{GS}-V_T$

$$V_{DS} = V_{DS}(sat) = V_{GS} - V_T \quad (11.16)$$

$$I_D(sat) = WC_{ox}\left(V_{GS} - V_T\right)v_{sat} \quad (11.17)$$

$$\mu = \frac{\mu_{eff}}{\left[1+\left(\frac{\mu_{eff}E}{v_{sat}}\right)^2\right]^{\frac{1}{2}}} \quad (11.18)$$

$$g_{ms} = \frac{\partial I_D(sat)}{\partial V_{GS}} = WC_{ox}v_{sat} \quad (11.19)$$

$$f_T = \frac{g_m}{2\pi C_G} = \frac{WC_{ox}v_{sat}}{2\pi\left(C_{ox}WL\right)} = \frac{v_{sat}}{2\pi L} \quad (11.20)$$

11.1.5 Ballistic Transport

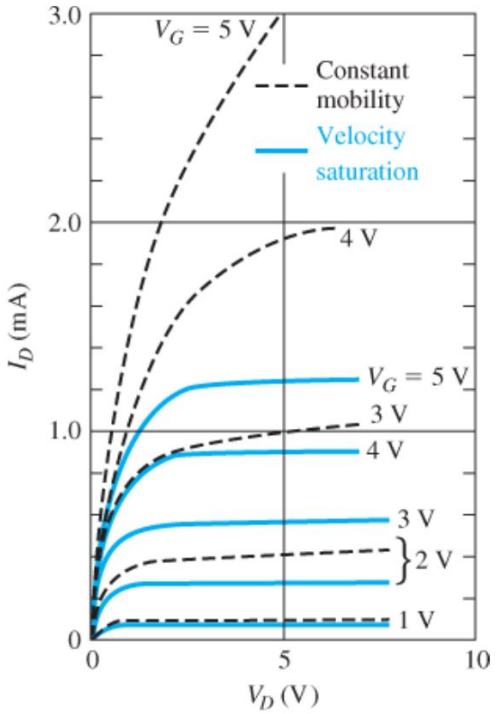


Figure 11.11 | Comparison of I_D versus V_D characteristics for constant mobility (dashed curves) and for field-dependent mobility and velocity saturation effects (solid curves).
(From Sze and Ng [22].)

11.2. 3 Constant Field Scaling

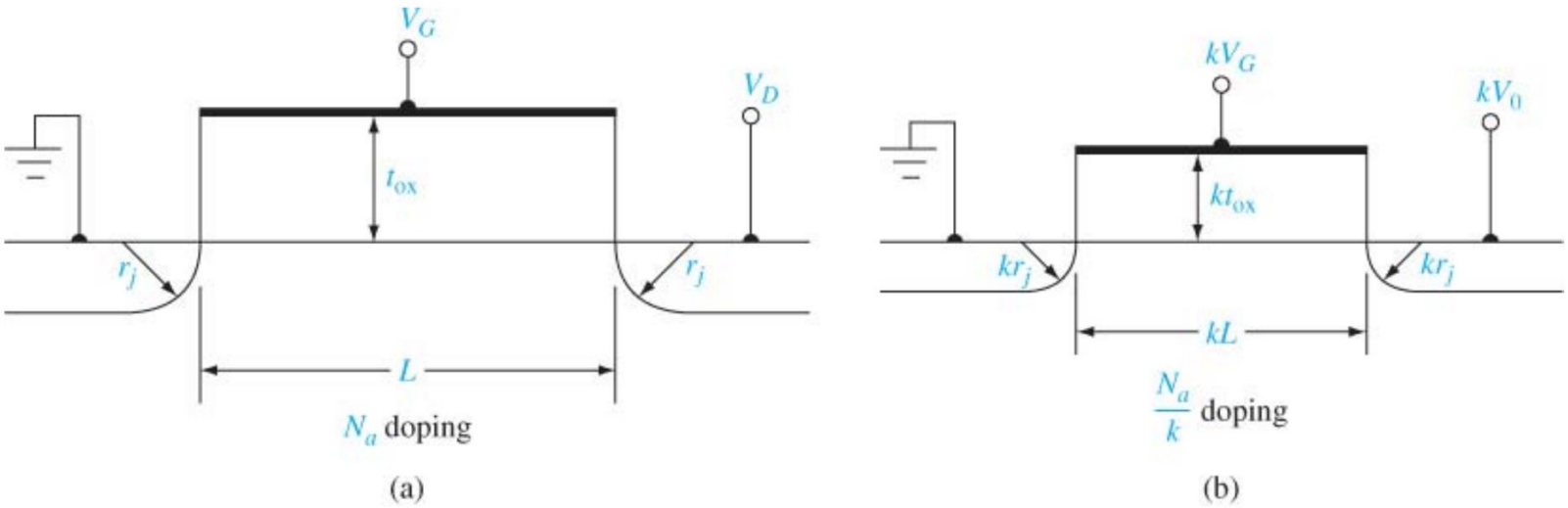


Figure 11.12 | Cross section of (a) original NMOS transistor and (b) scaled NMOS transistor.

11.2 MOSFET SCALING

Table 11.1 | Summary of constant-field device scaling

	Device and circuit parameters	Scaling factor ($k < 1$)
Scaled parameters	Device dimensions (L, t_{ox}, W, x_j)	k
	Doping concentration (N_a, N_d)	$1/k$
	Voltages	k
Effect on device parameters	Electric field	1
	Carrier velocity	1
	Depletion widths	k
	Capacitance ($C = \epsilon A/t$)	k
	Drift current	k
Effect on circuit parameters	Device density	$1/k^2$
	Power density	1
	Power dissipation per device ($P = IV$)	k^2
	Circuit delay time ($\approx CV/I$)	k
	Power−delay product ($P\tau$)	k^3

Source: Taur and Ning [23].

11.3 Threshold voltage modification

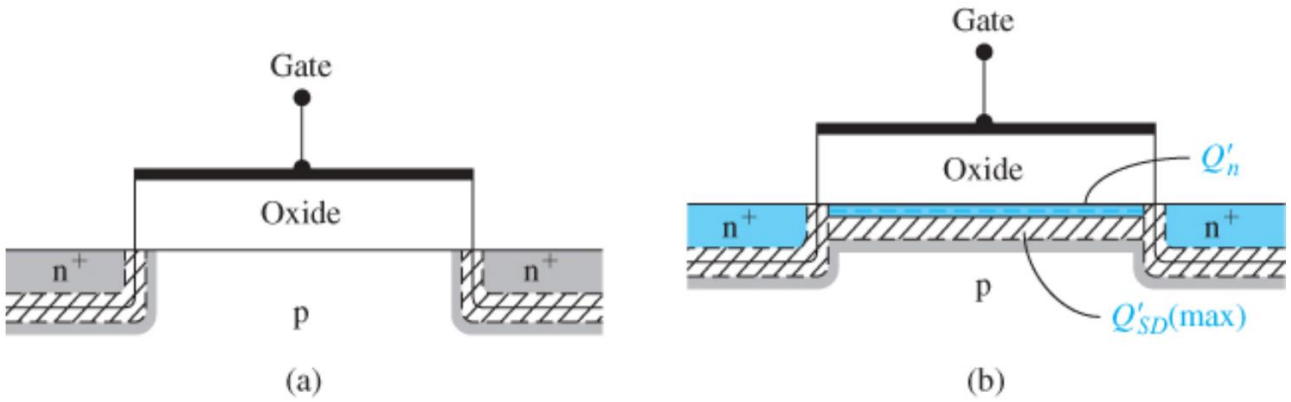


Figure 11.13 | Cross section of a long n-channel MOSFET (a) at flat band and (b) at inversion.

11.3 Threshold voltage modification

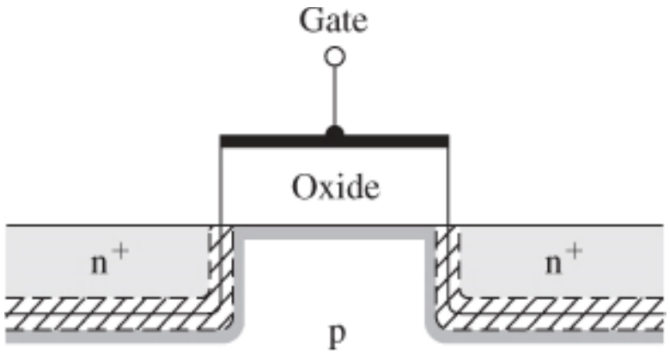


Figure 11.14 | Cross section of a short n-channel MOSFET at flat band.

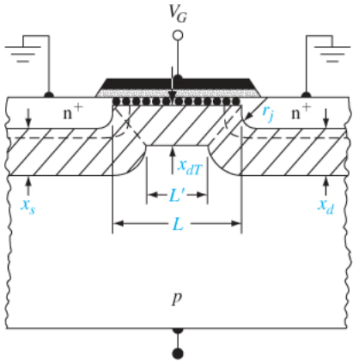


Figure 11.15 | Charge sharing in the short-channel threshold voltage model. *(From Yau [26].)*

$$x_D = \sqrt{\frac{2\varepsilon(V_{bi}+V_D)}{eN_a}} \quad (11.21)$$

$$\frac{I_D}{W} = \frac{\mu_n \varepsilon_{ox}}{2t_{ox}L}(V_G - V_T)^2 \rightarrow \frac{\mu_n \varepsilon_{ox}}{2(kt_{ox})(kL)}(kV_G - V_T)^2 \approx cons\tan t \quad (11.22)$$

$$V_T = V_{FB} + 2\phi_{fp} + \frac{\sqrt{2\varepsilon e N_a(2\phi_{fp})}}{C_{ox}} \quad (11.23)$$

11.3 Threshold voltage modification

$$V_{TN} = \left(\left|Q'_{SD}(\max)\right| - Q'_{ss}\right)\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp} \quad (11.24)$$

$$x_s \approx x_d \approx x_{dT} \equiv x_{dT} \quad (11.25)$$

$$\left|Q'_B\right| \cdot L = eN_a x_{dT}\left(\frac{L + L'}{2}\right) \quad (11.26)$$

$$\Delta V_T = V_{T(shortchannel)} - V_{T(longchannel)} \quad (11.30)$$

$$\frac{L + L'}{2L} = \left[1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.27)$$

$$\left|Q'_B\right| = eN_a x_{dT}\left[1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.28)$$

$$\Delta V_T = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.29)$$

11.3 Threshold voltage modification

$$V_{TN} = \left(\left|Q'_{SD}(\max)\right| - Q'_{ss}\right)\left(\frac{t_{ox}}{\varepsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp} \quad (11.24)$$

$$x_s \approx x_d \approx x_{dT} \equiv x_{dT} \quad (11.25)$$

$$\left|Q'_B\right| \cdot L = eN_a x_{dT}\left(\frac{L + L'}{2}\right) \quad (11.26)$$

$$\Delta V_T = V_{T(shortchannel)} - V_{T(longchannel)} \quad (11.30)$$

$$\frac{L + L'}{2L} = \left[1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.27)$$

$$\left|Q'_B\right| = eN_a x_{dT}\left[1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.28)$$

$$\Delta V_T = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] \quad (11.29)$$

Example 11.3(1)

① **Calculate the threshold voltage shift due to short-channel effects.**
- **N_a=3×10^{16} cm^{-3}, L=1.0 μm, r_j=0.3 μm, t_{ox}=20 nm=200 Å**

We can find

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85\times10^{-14})}{200\times10^{-8}} = 1.726\times10^{-7}\, F/cm^2$$

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{ni}\right) = (0.0259)\ln\left(\frac{3\times10^{16}}{1.5\times10^{10}}\right) = 0.3758V$$

and

$$\chi_{dT} = \left[\frac{4\varepsilon_s\phi_{fp}}{eN_a}\right]^{1/2} = \left[\frac{4(11.7)(8.85\times10^{-14})(0.3758)}{(1.6\times10^{-19})(3\times10^{-16})}\right]^{1/2}$$

$$= 0.18\times10^{-4}\, cm = 0.18\,\mu m$$

Example 11.3(2)

① **Calculate the threshold voltage shift due to short-channel effects.**

- **N_a=3×10^{16} cm^{-3}, L=1.0 μm, r_j=0.3 μm, t_{ox}=20 nm=200 Å**

The shift in threshold voltage is now found as

$$\Delta V_T = -\frac{eN_a\chi_{dT}}{C_{ox}}\left[\frac{r_j}{L}\left(\sqrt{1+\frac{2\chi_{dT}}{r_j}}-1\right)\right]$$

$$= -\frac{(1.6\times10^{-19})(3\times10^{16})(0.18\times10^{-4})}{1.726\times10^{-7}}\left[\frac{0.3}{1.0}\left(\sqrt{1+\frac{2(0.18)}{0.3}}-1\right)\right]$$

or

$$\Delta V_T = -0.0726V$$

- **If the design value of the threshold voltage of an n-channel MOSFET is to be V_T=0.35 V, for example, a shift of ΔV_T=−0.0726 V due to short-channel effects is significant and needs to be taken into account.**

11.3.2 Narrow-Channel Effects

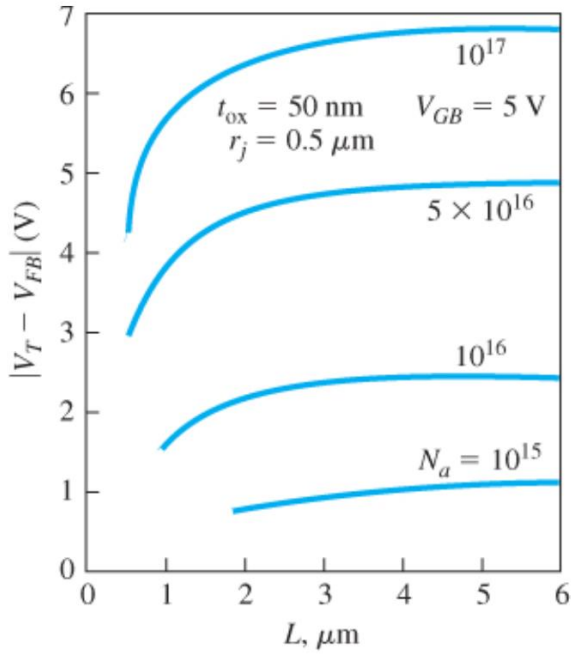


Figure 11.16 | Threshold voltage versus channel length for various substrate dopings. (*From Yau [26].*)

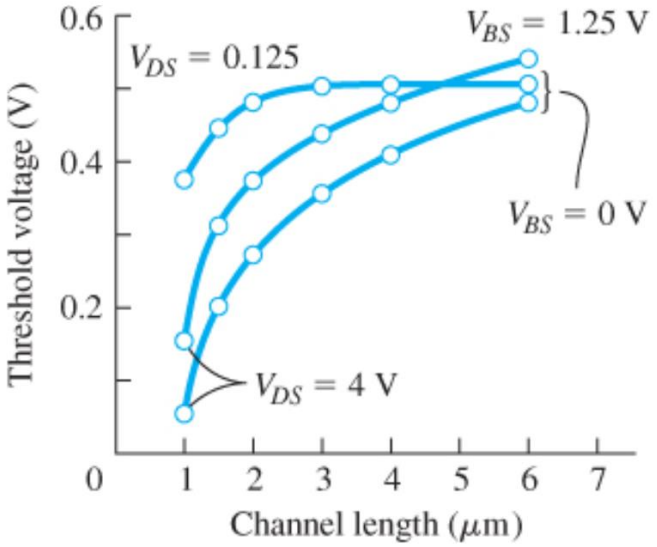


Figure 11.17 | Threshold voltage versus channel length for two values of drain-to-source and body-to-source voltage. (*From Yang [25].*)

Example 11.4

① **Design the channel width that will limit the threshold voltage shift because of narrow channel effects to a specified value.**

- **N_a=3×10^{16} cm^{-3}, t_{ox}=20 nm=200 Å, ξ=π/2, ΔV_T=0.2 V**

We find

$$C_{ox} = 1.726\times10^{-7}\, F/cm^2 \text{ and } \chi_{dT} = 0.18\mu m$$

From Equation (11.35), we can express the channel width as

$$W = \frac{eN_a\left(\xi x^2 dT\right)}{C_{ox}(\Delta VT)} = \frac{(1.6\times10^{-19})(3\times10^{16})\left(\frac{\pi}{2}\right)(0.18\times10^{-4})^2}{(1.726\times10^{-7})(0.2)}$$

$$= 7.08\times10^{-5}\, cm$$

$$W = 0.708\mu m$$

- **We may note that the threshold voltage shift of ΔV_T=0.2 V occurs at a channel width of W=0.708 μm, which is approximately four times larger than the induced space charge width x_{dT}.**

11.3.2 Narrow-Channel Effects

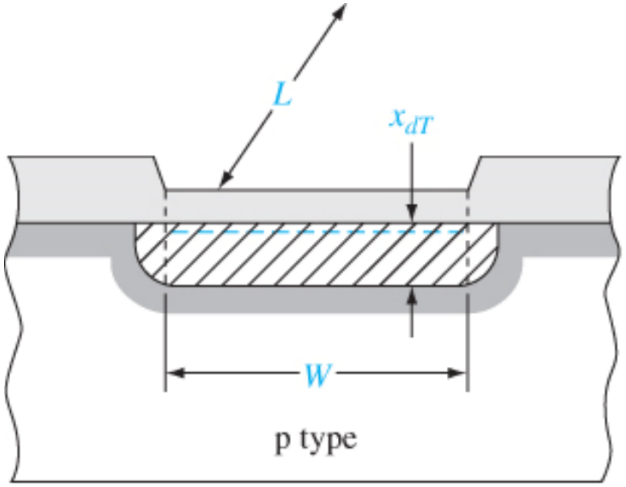


Figure 11.18 | Cross section of an n-channel MOSFET showing the depletion region along the width of the device.

$$Q_B = Q_{B0} + \Delta Q_B \quad (11.31)$$

$$|Q_{B0}| = eN_a WLx_{dT} \quad (11.32)$$

$$\Delta Q_B = eN_a Lx_{dT}(\xi x_{dT}) \quad (11.33)$$

$$|Q_B| = |Q_{B0}| + |\Delta Q_B| = eN_a WLx_{dT} + eN_a Lx_{dT}(\xi x_{dT}) = eN_a WLx_{dT}\left(1 + \frac{\xi x_{dT}}{W}\right) \quad (11.34)$$

11.3.2 Narrow-Channel Effects

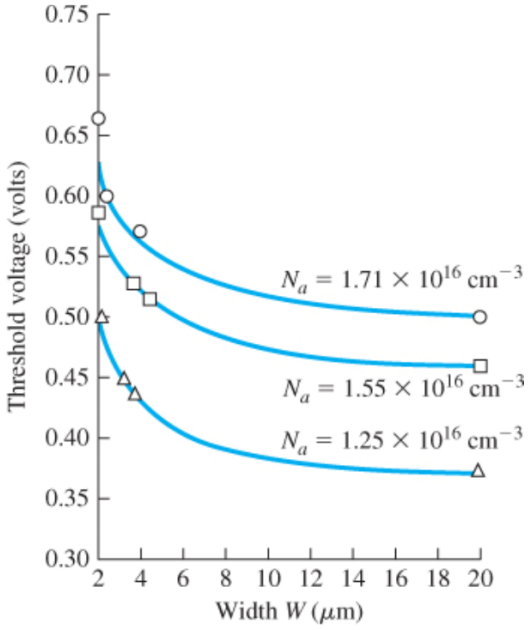


Figure 11.19 | Threshold voltage versus channel width (solid curves, theoretical; points, experimental).
(*From Akers and Sanchez [1].*)

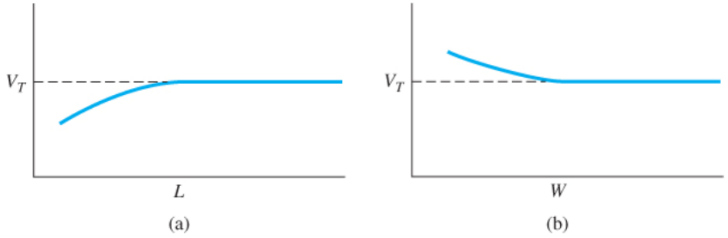


Figure 11.20 | Qualitative variation of threshold voltage (a) with channel length and (b) with channel width.

11.4. Additional Electrical Characteristics

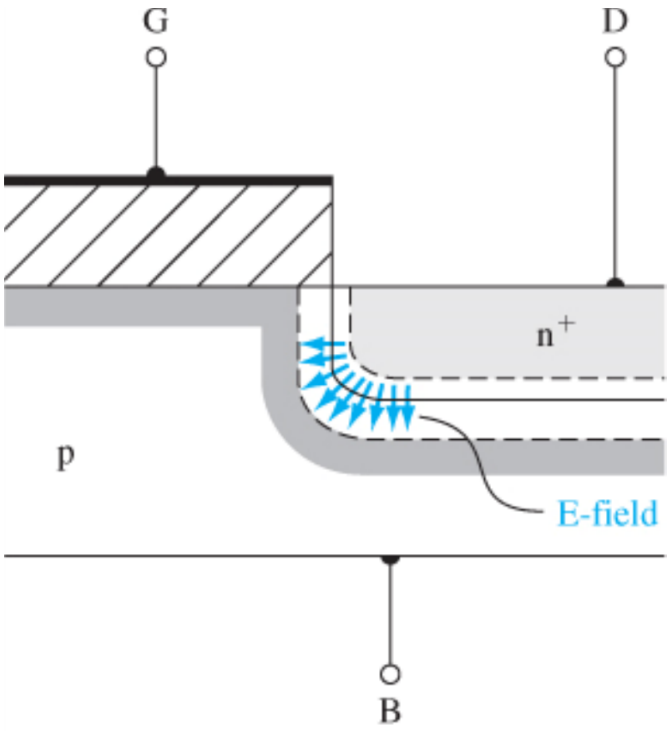


Figure 11.21 | Curvature effect on the electric field in the drain junction.

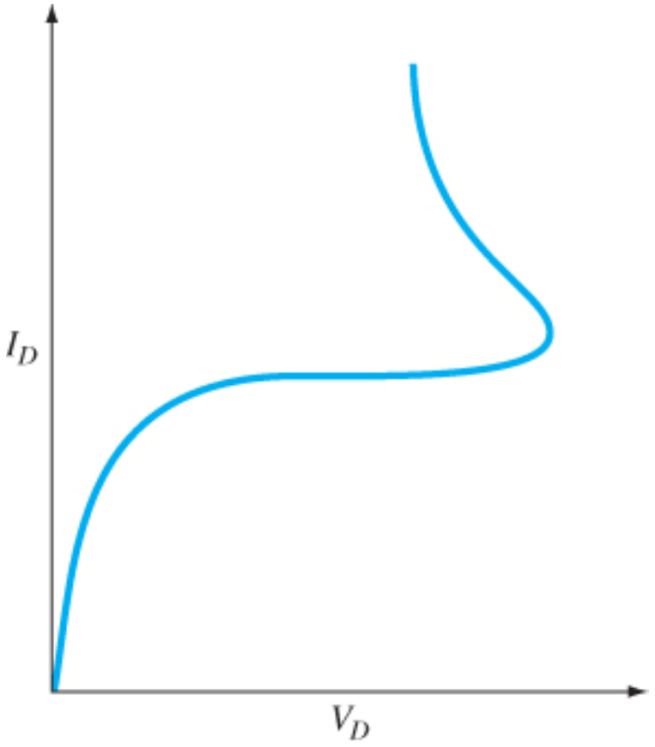


Figure 11.22 | Current–voltage characteristic showing the snapback breakdown effect.

11.4. Additional Electrical Characteristics

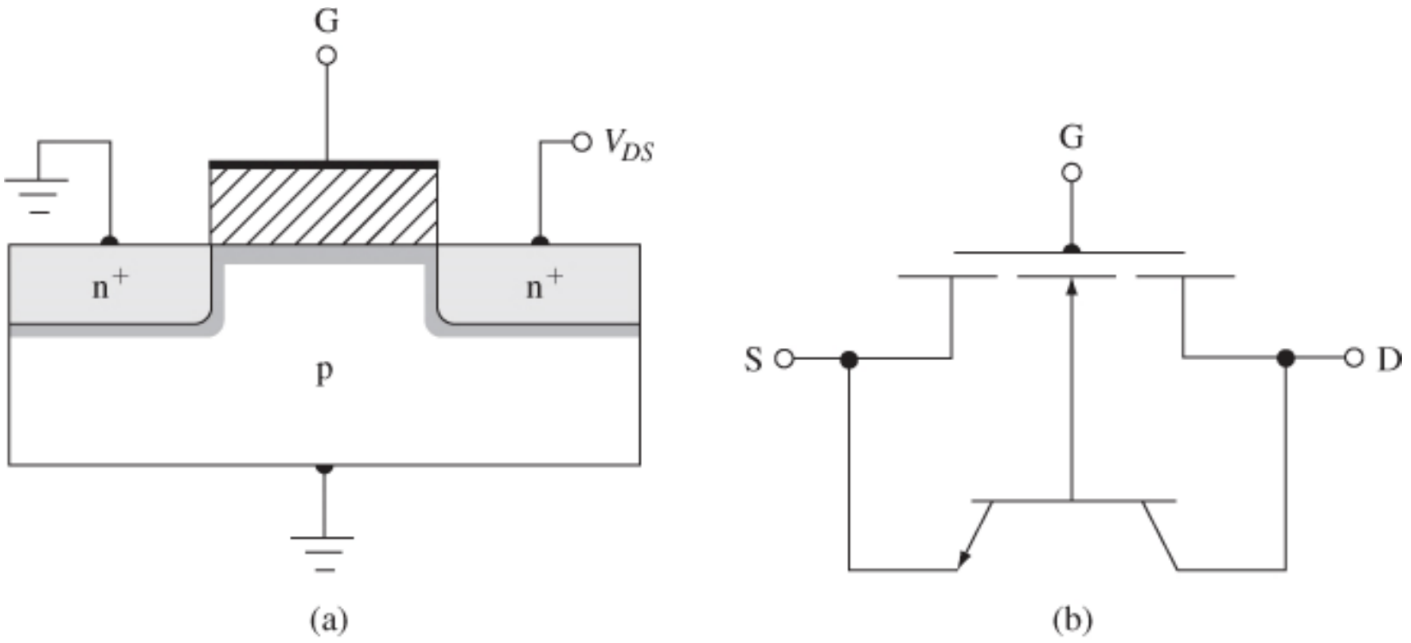


Figure 11.23 | (a) Cross section of the n-channel MOSFET. (b) Equivalent circuit including the parasitic bipolar transistor.

11.4. Additional Electrical Characteristics

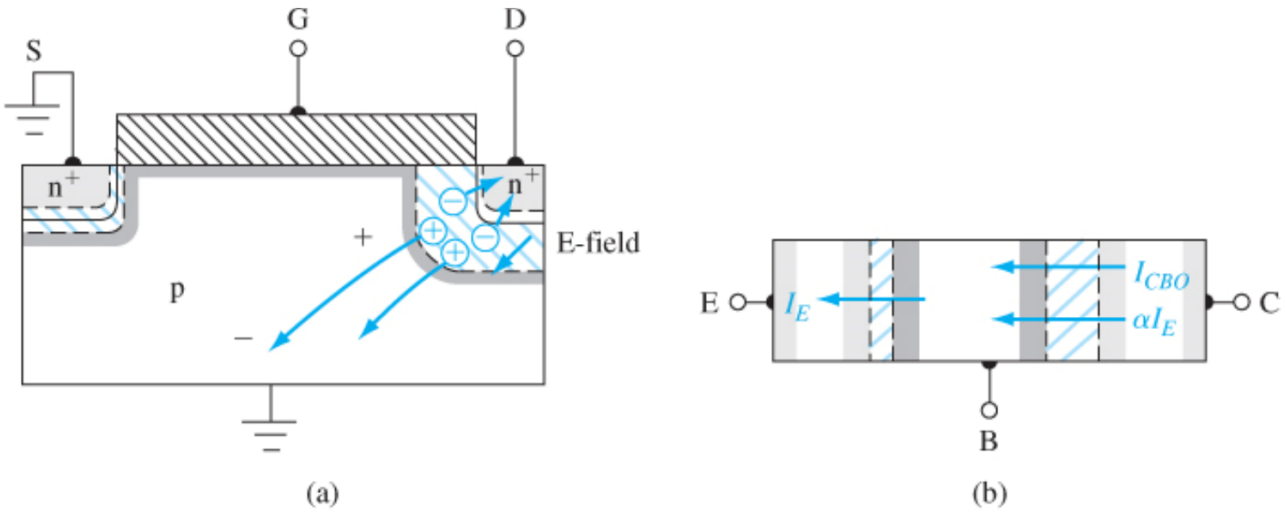


Figure 11.24 | (a) Substrate current–induced voltage drop caused by avalanche multiplication at the drain. (b) Currents in the parasitic bipolar transistor.

11.4.1 Breakdown Voltage

$$I_C = \alpha I_E + I_{CB0} \quad (11.36)$$

$$I_C = \alpha I_C + I_{CB0} \quad (11.37)$$

$$I_C = M\left(\alpha I_C + I_{CB0}\right) \quad (11.38)$$

$$I_C = \frac{MI_{CB0}}{1-\alpha M} \quad (11.39)$$

$$M = \frac{1}{1-\left(\frac{V_{CE}}{V_{BD}}\right)^m} \quad (11.40)$$

11.4. Near punch-through Effects

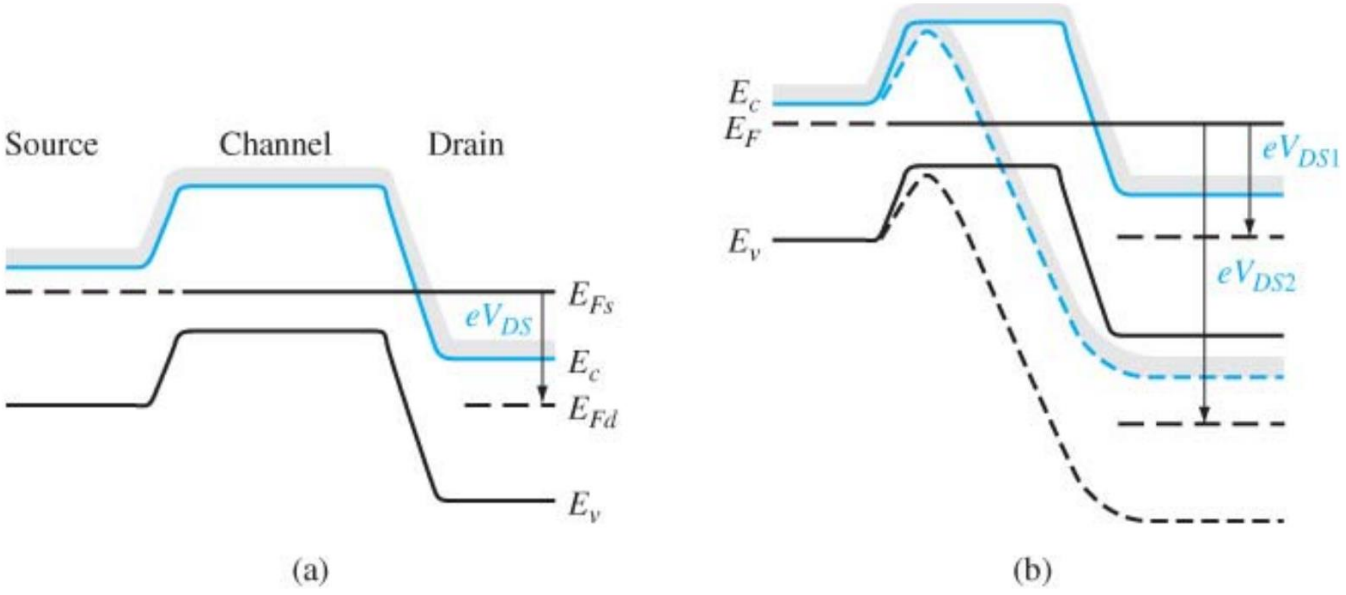


Figure 11.25 | (a) Equipotential plot along the surface of a long-channel MOSFET. (b) Equipotential plot along the surface of a short-channel MOSFET before and after punch-through.

11.4. Near punch-through Effects

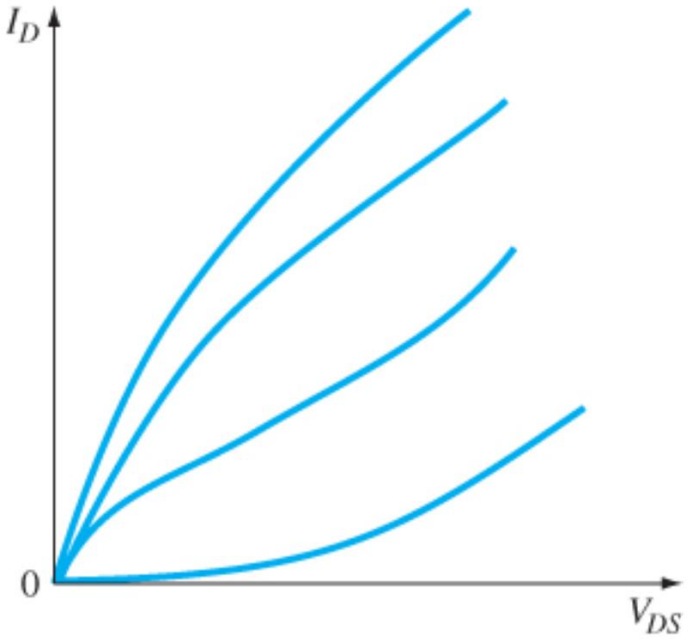


Figure 11.26 | Typical *I–V* characteristics of a MOSFET exhibiting punch-through effects.

Example 11.5(1)

① **Calculate the theoretical punch-through voltage assuming the abrupt junction approximation.**
- **$N_d=10^{19}$ cm^{-3}, $N_a=10^{16}$ cm^{-3}, L=1.2 μm**

The pn junction built-in potential barrier is given by

$$V_{bi} = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.0259)\ln\left[\frac{(10^{16})(10^{19})}{(1.5\times10^{10})^2}\right] = 0.874V$$

The zero-biased source-substrate pn junction width is

$$\chi_{d0} = \left[\frac{2\varepsilon_s V_{bi}}{eN_a}\right]^{1/2} = \left[\frac{2(11.7)(8.85\times10^{-14})(0.874)}{(1.6\times10^{-19})(10^{16})}\right]^{1/2} = 0.336\mu m$$

The reverse-biased drain-substrate pn junction width is given by

$$\chi_d = \left[\frac{2\varepsilon_s(V_{bi}+V_{DS})}{eN_a}\right]^{1/2}$$

At punch-through, we will have

$$\chi_{d0} + \chi_d = L \text{ or } 0.336 + \chi_d = 1.2$$

Example 11.5(2)

① **Calculate the theoretical punch-through voltage assuming the abrupt junction approximation.**
- **$N_d=10^{19}$ cm^{-3}, $N_a=10^{16}$ cm^{-3}, L=1.2 μm**

which gives $\chi_d = 0.864\mu$m at the punch - through condition. We can then find

$$V_{bi}+V_{DS}=\frac{\chi_d{}^2 eN_a}{2\varepsilon_s}=\frac{(0.864\times10^{-4})^2(1.6\times10^{-19})(10^{16})}{2(11.7)(8.85\times10^{-14})}$$

$$=5.77V$$

The punch - through voltage is then found as

$$V_{DS}=5.77-0.874=4.9V$$

- **As the two space charge regions approach punch-through, the abrupt junction approximation is no longer a good assumption. The drain current will begin to increase rapidly before the theoretical punch-through voltage is reached.**

11.4.2 The lightly Doped Drain Transistor

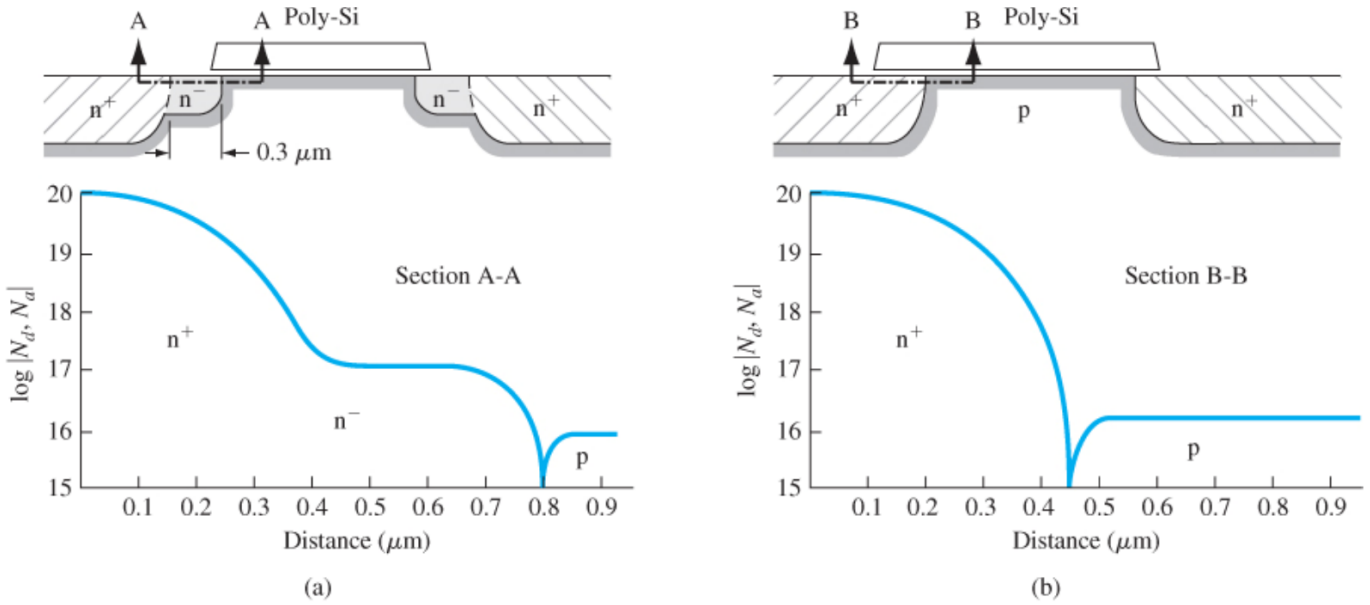


Figure 11.27 | (a) The lightly doped drain (LDD) structure. (b) Conventional structure. (*From Ogura et al. [12].*)

11.4.2 The lightly Doped Drain Transistor

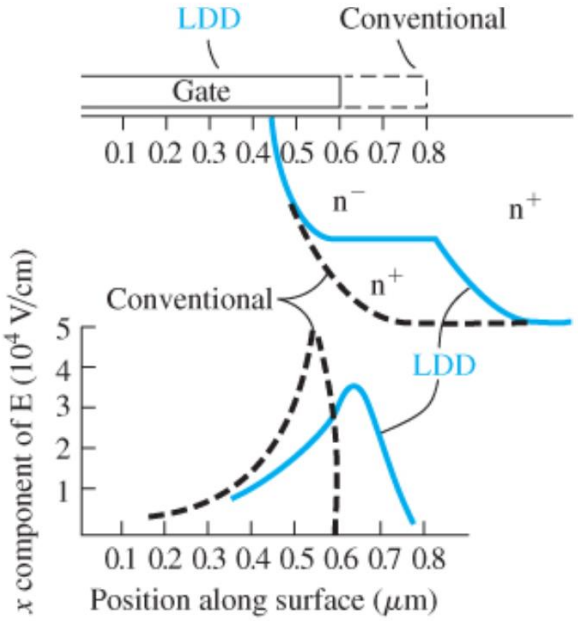


Figure 11.28 | Magnitude of the electric field at the Si–SiO_2 interface as a function of distance; $V_{DS} = 10$ V. $V_{SB} = 2$ V, $V_{GS} = V_T$. (*From Ogura et al. [12].*)

11.4.3 Threshold Adjustment by Ion Implantation

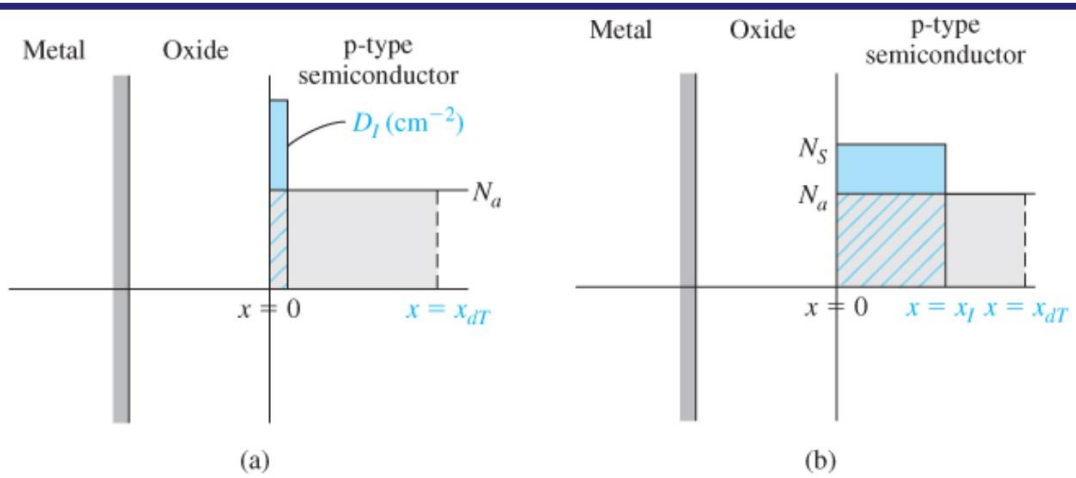


Figure 11.29 | (a) Ion-implanted profile approximated by a delta function. (b) Ion-implanted profile approximated by a step function in which the depth x_i is less than the space charge width x_{dT}.

$$\Delta V_T = +\frac{eD_1}{C_{ox}} \quad (11.41)$$

$$x_{dT} = \sqrt{\frac{2\varepsilon s}{eN_a}\left[2\phi_{fp} - \frac{ex_1^2}{2\varepsilon_s}\left(N_s - N_a\right)\right]^{\frac{1}{2}}} \quad (11.42)$$

$$V_T = V_{T0} + \frac{eD_1}{C_{ox}} \quad (11.43)$$

$$D_1 = \left(N_s - N_a\right)x_1 \quad (11.44)$$

$$V_{T0} = V_{FB0} + 2\phi_{fp0} + \frac{eN_a x_{dT0}}{C_{ox}} \quad (11.45)$$

Example 11.6(1)

① **Design the ion implant dose required to adjust the threshold voltage to a specified value.**
- **N_a=5×10^{15} cm^{-3}, t_{ox}=18 nm=180 Å, V_{FBO}=−1.25 V, V_T=0.4 V**

We find that

$$\phi_{fp0} = V_t \ln\left(\frac{N_a}{n_i}\right) = (0.0259)\ln\left(\frac{5\times10^{15}}{1.5\times10^{10}}\right) = 0.3294V$$

$$\chi_{dT0} = \left[\frac{4\varepsilon_s\phi_{fp0}}{eN_a}\right]^{1/2} = \left[\frac{4(11.7)(8.85\times10^{-14})(0.3294)}{(1.6\times10^{-19})(5\times10^{15})}\right]^{1/2}$$

$$= 0.4130\times10^{-4}\,cm$$

and

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85\times10^{-14})}{180\times10^{-8}} = 1.9175\times10^{-7}\,F/cm^2$$

The initial pre − implant threshold voltage is

$$V_{TO} = V_{FBO} + 2\phi_{fp0} + \frac{eN_a\chi_{dTO}}{C_{ox}}$$

$$= -1.25 + 2(0.3294) + \frac{(1.6\times10^{-19})(5\times10^{15})(0.4130\times10^{-4})}{1.9175\times10^{-7}}$$

$$= -0.419V$$

Example 11.6(2)

① **Design the ion implant dose required to adjust the threshold voltage to a specified value.**
- **N_a=5×10^{15} cm^{-3}, t_{ox}=18 nm=180 Å, V_{FBO}=−1.25 V, V_T=0.4 V**

The threshold voltage after implant, from Equation (11.43), is

$$V_T = V_{TO} + \frac{eD_I}{C_{ox}}$$

so that

$$+0.40 = -0.419 + \frac{(1.6\times10^{-19})D_I}{1.9175\times10^{-7}}$$

which gives

$$D_I = 9.815\times10^{-11} cm^{-2}$$

If the uniform step implant extends to a depth of $x_I = 0.15\mu$m, for example, then the equivalent acceptor concentration at the surface is

Example 11.6(3)

① Design the ion implant dose required to adjust the threshold voltage to a specified value.
- $N_a=5\times10^{15}$ cm^{-3}, $t_{ox}=18$ nm$=180$ Å, $V_{FBO}=-1.25$ V, $V_T=0.4$ V

$$N_s - N_a = \frac{D_I}{x_I}$$

or

$$N_s - 5\times10^{15} = \frac{9.815\times10^{11}}{0.15\times10^{-4}}$$

so that

$$N_s = 7.04\times10^{16}\,cm^{-3}$$

- It is assumed in the above calculation that the induced space charge width in the channel region is greater than the ion implant depth x_I. The calculation satisfies our assumptions.

11.5 Radiation And Hot-Electron Effects

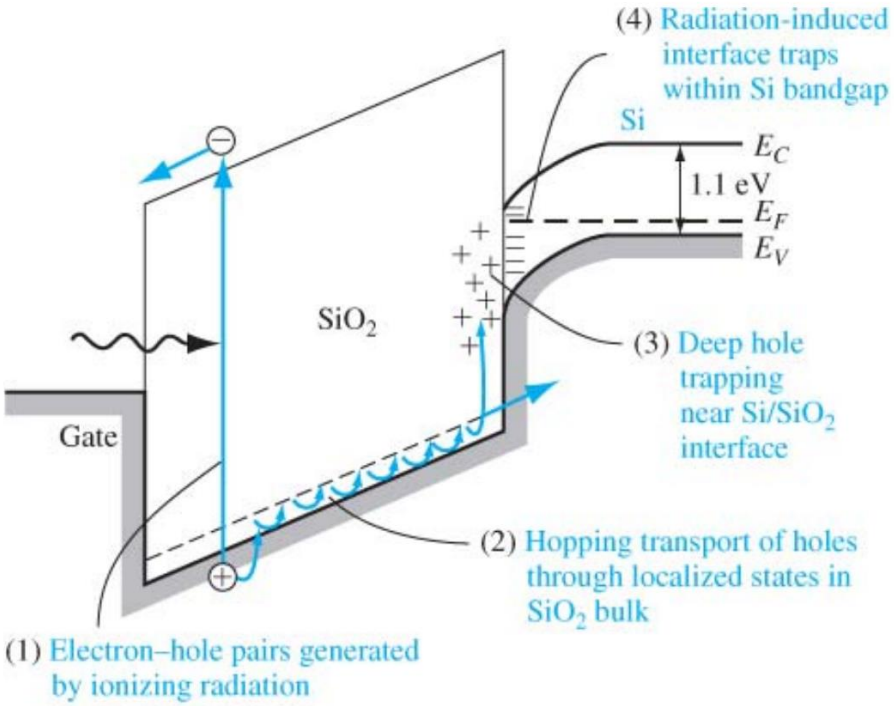


Figure 11.30 | Schematic of ionizing radiation–induced processes in an MOS capacitor with a positive gate bias. (*From Ma and Dressendorfer [7].*)

11.5 Radiation And Hot-Electron Effects

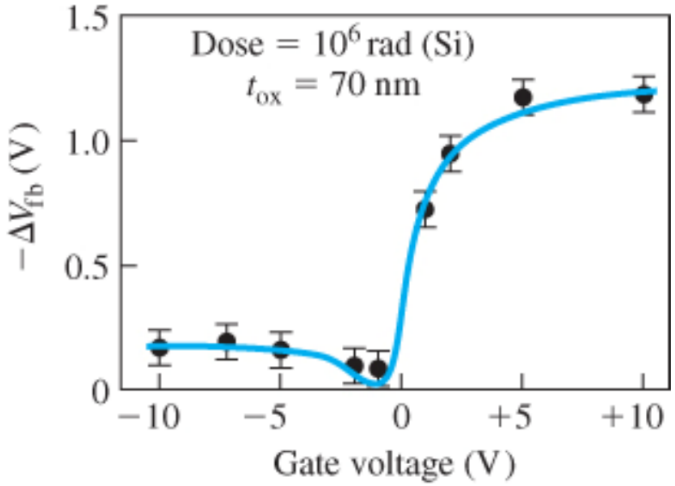


Figure 11.31 | Radiation-induced flat-band voltage shift in an MOS capacitor as a function of applied gate bias during irradiation.
(From Ma and Dressendorfer [7].)

Example 11.7

① **Calculate the threshold voltage shift due to radiation-induced oxide charge trapping.**

- **t_{ox}=25 nm=250 Å**
- **Assume that a pulse of ionizing radiation creates 10^{18} electron-hole pairs per cm^3**
- **Assume that the electrons are swept out through the gate terminal with zero recombination, and that 20 percent of the generated holes are trapped at the oxide-semiconductor interface.**

The areal density of holes generated in the oxide is

$$N_h = (10^{18})(250\times10^{-8}) = 2.5\times10^{12}\,cm^{-2}$$

The equivalent trapped surface charg is then

$$Q'_{ss} = (2.5\times10^{12})(0.2)(1.6\times10^{-19}) = 8\times10^{-8}\,C/cm^2$$

We find

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85\times10^{-14})}{250\times10^{-8}} = 1.381\times10^{-7}\,F/cm^2$$

The threshold voltage shift is then

$$\Delta V_T = -\frac{Q'_{ss}}{C_{ox}} = -\frac{8\times10^{-8}}{1.381\times10^{-7}} = -0.579V$$

- **As we have seen previously, a positive fixed oxide charge shifts the threshold voltage in the negative voltage direction. The ionizing radiation may shift an enhancement-mode device into the depletion mode.**

11.5.2 Radiation-induced Interface States

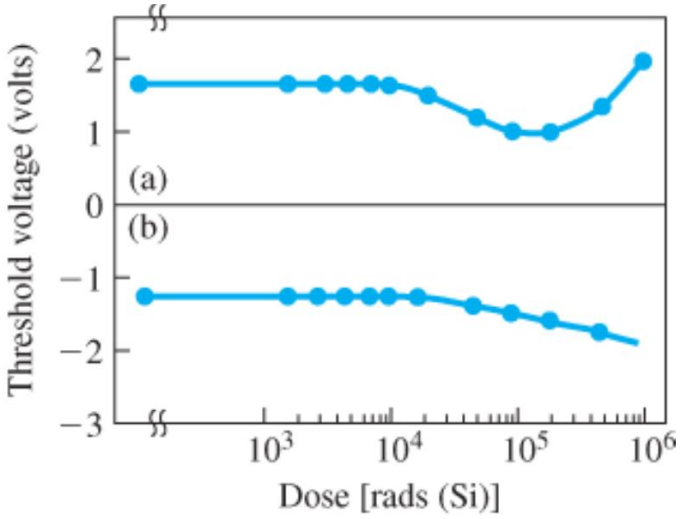


Figure 11.32 | Threshold voltage versus total ionizing radiation dose of (a) an n-channel MOSFET and (b) a p-channel MOSFET.
(From Ma and Dressendorfer [7].)

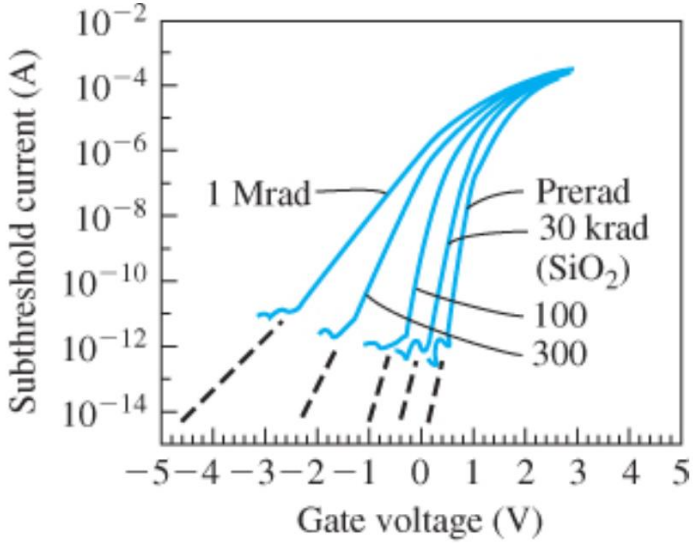


Figure 11.33 | Subthreshold current versus gate voltage of a MOSFET prior to irradiation and at four total radiation dose levels.
(From Ma and Dressendorfer [7].)

11.5.2 Radiation-induced Interface States

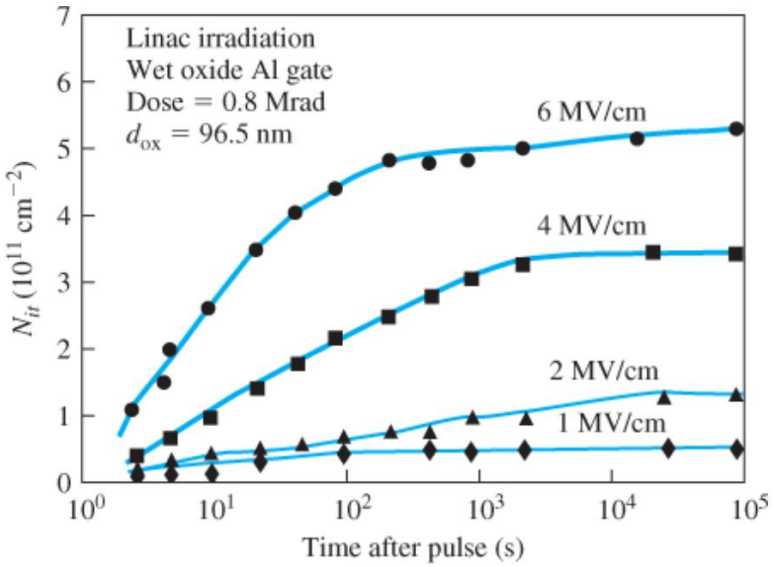


Figure 11.34 | Radiation-induced interface state density versus time after a pulse of ionizing radiation for several values of oxide electric field.
(From Ma and Dressendorfer [7].)

11.5.3 Hot-Electron Charging Effects

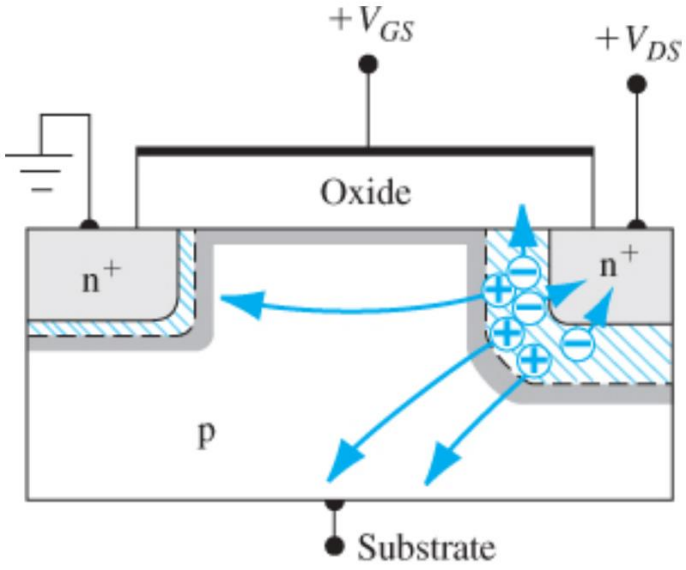


Figure 11.35 | Hot carrier generation, current components, and electron injection into the oxide.